

Handpiece Integrating Tactile Feedback for Surgical VR-Training of Total Hips Arthroplasty (THA)

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ABSTRACT

Surgical Virtual Reality training systems for Total Hips Arthroplasty (THA) with haptic feedback are rare. Here, we present the development of a handpiece for such a simulator usable for two steps of THA. The handpiece was 3D-printed from metal and incorporates microelectronics that allow it to vibrate, detect button presses and communicate via WIFI. As part of a larger study, the realism of the handpiece was evaluated and generally confirmed, although the vibrations emitted were too low. Using shakers and the haptic device it is connected to, might solve this issue.

Index Terms: Applied computing → Life and medical sciences, Human-centered computing → Virtual reality, Human-centered computing → Haptic devices

1 INTRODUCTION

Surgical training in Virtual Reality (VR) is a very important use case to reduce training time on real patients [1]. Especially, realistic haptic and tactile feedback is mandatory for surgeons to acquire skills. Despite the many existing haptic VR surgical training simulators, applications for Total Hips Arthroplasty (THA – hip joint implants), barely exist [2, 3]. For the surgical steps of cutting off the femoral head and reaming the acetabulum a surgical saw and a surgical reamer are used. Depending on the surgeons' preferences and the implant type the saw and reamer vary, but can be categorized in electrical or pneumatical driven devices. Most of those device's shapes resemble hand drilling machine and are operated with one or two buttons. During operation, the surgical instruments emit vibrations. So far, only one work by Weik et al. [4] reports on integrating tactile feedback in an original reaming device used for hip pan reaming. We aimed at designing a common tactile handpiece for both reaming and sawing, that can emit the vibrations felt during really surgery. Further, it should be shaped and operated like real surgical tools. This handpiece was used and evaluated as a part of a VR simulator using haptic feedback for learning THA where the study design is described [3].

2 MECHANICAL DESIGN AND MANUFACTURING OF THE REAMER

The handpiece transmits forces and torques from the haptic device that it is connected to and the tactile vibration from the handpiece itself (Figure 1, green). It is connectable with two haptic devices

(Vituose devices, Haption, and KUKA iiwa series; Figure 1, yellow) and allows the mounting of different rods (Figure 1, red). Its shape resembles real surgical instruments, which have a cylindrical contour in the area of the tool holder. Various electronic components can be installed in this cylinder. For this purpose, various grooves were provided in which assembly components could be clamped (Figure 2). Attached to the cylinder is a handle, which has an oval contour to allow an ergonomic grip. The two knobs on the handle allow the user to control the drilling direction and speed. It consists of two half-shells to assemble the electronic components. An initial Finite Element Analysis with various plastic materials at an axial force of 140 N resulted in a displacement in the joining zones of up to 1.78 mm. For this reason, selective laser melting was used to manufacture the handpiece from aluminum. 140 N was chosen as load as this is the maximum force of the KUKA iiwa 14 R820, which is the most powerful haptic device the handpiece is attached to and because such forces can occur during acetabulum reaming [5].

3 MICROELECTRONIC DESIGN AND IMPLEMENTATION

3.1 Requirements

Vibrotactile feedback was to be integrated into the reamer to simulate the vibrations that occur during the reaming and sawing. Resembling the real surgical instruments, two buttons were used as manual triggers and the electronics had to be fully integrated into the interior of the reamer. In addition, the intensity of the vibration had to be clearly perceptible and variable. We wanted a maintenance-friendly system with an on/off switch, battery operation, visual feedback (e.g. battery charge status) and a wireless interface (Bluetooth or WiFi).

3.2 Circuit design, component selection and properties

A circuit design for the vibrotactile feedback was developed based on the requirement specifications and the available installation space. Figure 2 shows the assembly of all electronic components inside the handpiece. An Arduino Nano 33 IOT serves as the system's central computing and control unit, which was selected in particular due to its small dimensions, wireless interfaces and established programming environment.

The system is powered by a 3.7 V 4400 mAh Li-Ion battery and a Pololu S7V8F5 5 V step-up/step-down converter with integrated overheat and short-circuit protection. For the charging process, a separate circuit board (Adafruit Micro-Lipo Charger with Micro-USB Jack) was installed in the rear area of the reamer to ensure good accessibility to the USB port. The battery charge status is determined by an implemented voltage divider with two resistors (330 kΩ and 1.2 MΩ) and a capacitor (100 nF). Two low-current LEDs at the bottom of the circuit board serve as visual feedback. A transistor circuit (CENTRAL SEMICONDUCTOR 2N2222A) was implemented to control the vibration motors and simultaneously protect the Arduino pins from excessive currents, which means that the Arduino only has to provide the low base current to

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activate the motors. Furthermore, two 1N4001 flyback diodes and two 0.1 μF capacitors were added to protect the system from back EMF. To generate the vibrations, a Vybronic VJQ24-35F580C with a large flywheel mass and a Vybronic VJQ24-35K270B with a small flywheel mass were integrated. The vibration is activated via two additively manufactured buttons, each of which is coupled to a 10 k Ω potentiometric displacement sensor (MEGATRON, MM(R)10-8).

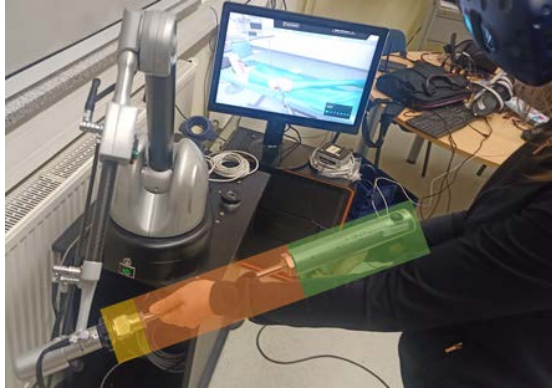


Figure 1: User operating the handpiece connected to haptic device (Virtuose 6D, Haption). Handpiece (green), connecting rod (red), adapter for Virtuose 6D (yellow)

3.3 Assembly and Mode of Operation

All electrical components were connected via a perfboard (2.54 pitch), which was manually fitted with conductors and connectors (JST XH, socket connectors). This enables quick maintenance in the event of a defective component. Furthermore, insulating adhesive tape was applied to prevent short circuits due to the metal housing. On the hardware side, three modes were implemented, that can be operated via an SPST rocker switch:

- "On" (I): The vibration board is powered by the battery and the vibration can be activated by pressing a button.
- "Charging position" (O): The battery is only connected to the charging board and can be charged.
- "Off" (II): The battery is not connected to either the charging board or the vibration board.

3.4 WIFI-Communication Protocol

For realistic haptic feedback, the handpiece's button presses and the activation or deactivation of vibration do not have a strict real-time constraint. The standard <100 ms for UI is sufficient. Therefore, the communication between the handpiece and the haptic simulation was implemented as a simple UDP connection. This connection consists of one package sent from the device to the simulation containing the percentage of each button pressed as a float value between 0 & 1, each and a package from the simulation to the device containing a value for each vibration motor. These packages are sent in regular intervals with the last message always being the most up to date data. Any unattended messages in between are discarded. Additionally, the connection to the network is monitored and its status transmitted through blinking LEDs on the handpiece.

4 EVALUATION AND FURTHER DEVELOPMENT

In a study, described in [3], the realism of the VR training simulator, including the developed hand piece, was evaluated with 17 surgeons. In general, the handpiece fulfilled its cause allowing the participant to virtually cut off the femoral head and ream the acetabulum. However, the participants reported that the vibration

of the handpiece was far too soft compared to the real surgery. However, the most interesting result was, that 6 participants perceived the handpiece as too heavy, whilst others perceived it as realistic. The explanation for this is, that the real surgical instruments are powered either pneumatically or electrically, resulting in considerably different weights. For improving the intensity of the vibrations, it is not feasible to use more or larger vibration motors due to the limited space inside the handpiece. Integrating linear shaker internally hitting the handpiece could result in stronger vibrations. In addition, the haptic devices the hand piece is connected to should also transmit vibrations.

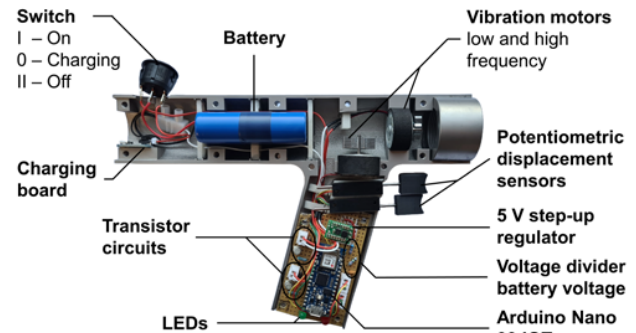


Figure 2: The handpiece showing all electronic components.

5 CONCLUSION

We presented the development of a handpiece integrating tactile feedback for a surgical VR training system for THA. The handpiece integrates two vibration motors, two potentiometers as buttons and an Arduino Nano. Via WIFI, the handpiece connected to the VR simulation. An evaluation with 17 surgeons showed that the handpiece is realistic but that the vibrations are too soft. An evolved version might use shakers and the haptic device to generate stronger vibrations.

ACKNOWLEDGEMENTS

The project on which this paper is based was funded by the German Federal Ministry of Education and Research under the funding code 16SV8356. The responsibility for the content of this publication lies with the authors.

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